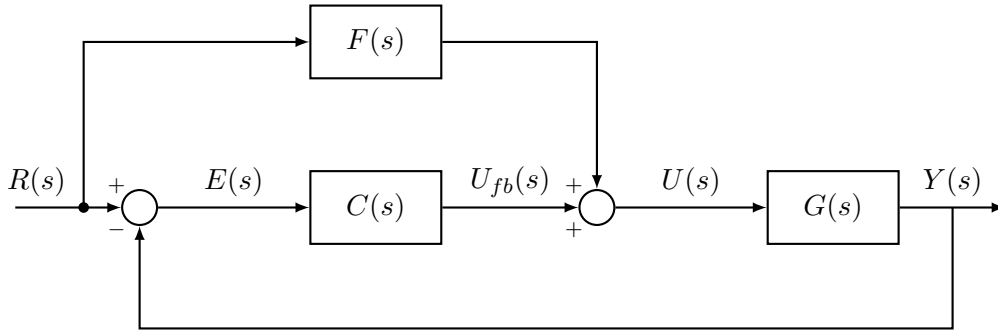


Block Diagram Reduction: Feedforward + Feedback

Block Diagram

The feedforward-augmented feedback loop is shown below. The reference $R(s)$ drives two paths simultaneously: the feedforward block $F(s)$ (direct) and the error-forming junction (via feedback).



Node Equations

Label each signal by inspecting every summing junction and block output:

$$E = R - Y \quad (1)$$

$$U_{fb} = C E \quad (2)$$

$$U = U_{fb} + F R \quad (3)$$

$$Y = G U \quad (4)$$

Algebraic Reduction

Substitute (1) into (2), then substitute both into (3):

$$U = C[R - Y] + F R.$$

Substitute into (4):

$$Y = G\{C[R - Y] + F R\}.$$

Expand:

$$Y = G C R - G C Y + G F R.$$

Collect all Y terms on the left:

$$Y[1 + GC] = G[C + F]R.$$

Divide both sides by $[1 + GC]$:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)[C(s) + F(s)]}{1 + G(s)C(s)}$$

Interpretation

The denominator $1 + G(s)C(s)$ is identical to the denominator of a pure feedback loop. Feedforward does not alter the characteristic equation and therefore does not change the closed-loop poles.

The numerator $G(s)[C(s) + F(s)]$ shows that the effective forward-path gain is the sum of the feedback controller and the feedforward controller. When the feedforward path is removed ($F(s) = 0$), the expression reduces to the familiar unity-feedback result:

$$T(s)|_{F=0} = \frac{G(s)C(s)}{1 + G(s)C(s)}.$$

Choosing F for Zero Steady-State Error

For a step reference, zero steady-state error requires $T(0) = 1$ (unity DC gain). Evaluating $T(s)$ at $s = 0$:

$$T(0) = \frac{G(0)[C(0) + F(0)]}{1 + G(0)C(0)} = 1.$$

Solving for $F(0)$:

$$\begin{aligned} G(0)[C(0) + F(0)] &= 1 + G(0)C(0) \\ G(0)F(0) &= 1 \\ F(0) &= \frac{1}{G(0)}. \end{aligned}$$

The DC gain of the feedforward block must equal the reciprocal of the plant's DC gain. Denoting the plant DC gain $K_{dc} = G(0)$, the condition is simply

$$F(0) = \frac{1}{K_{dc}}.$$

The simplest controller that satisfies this is the constant gain $F(s) = 1/K_{dc}$. This is the *proportional feedforward*: it scales the reference by the inverse of the plant's steady-state gain, delivering exactly the input the plant needs at steady state without waiting for error to accumulate.

Note that a feedback-only loop with an integrator ($C(s)$ containing $1/s$) also achieves $T(0) = 1$ by driving $C(0) \rightarrow \infty$. Feedforward achieves the same result without relying on infinite loop gain, which improves transient response and allows smaller feedback gains.